

Topic 05: Applications to Ordinary Differential Equations

Unit V — Inverse Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

Two full units. Twelve topics in Unit IV, five topics in Unit V. You have learned the definition, existence conditions, all standard transforms, five operational properties, derivatives, integrals, multiplication and division by t , evaluation of integrals, inverse transforms, partial fractions, and convolution. All of it has been building toward one destination: solving ordinary differential equations that arise in electrical circuits, mechanical systems, control engineering, and heat flow. This is that destination. In this topic, you will solve ODEs that would take pages using classical methods — each solved in under half a page using the Laplace Transform. Let's go.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this. The Laplace Transform method does something no classical technique can match: it converts an ODE together with its initial conditions into a pure algebraic equation in s . You solve for $Y(s)$, apply one of the four inversion methods you have now mastered, and the solution $y(t)$ appears directly — with initial conditions automatically absorbed.

More importantly, the method handles *discontinuous* and *impulsive* forcing functions — unit steps and Dirac deltas — with the same ease as smooth forcing. The table below shows what happens when students skip the foundations.

Mistake	Consequence
Skipping Units IV–V and going directly to the ODE solutions	No understanding of <i>why</i> each step works — just following a recipe
Forgetting to substitute initial conditions before inverting	Getting the wrong particular solution without ICs
Not recognising which inversion method to use	Stuck at $Y(s)$ — cannot recover $y(t)$

Key Insight

The Laplace Transform converts the IVP problem into algebra. Initial conditions are not patched on afterward — they are built into the algebraic equation from Step 1. This is the single greatest advantage of the method.

3 The Universal 4-Step Workflow

The Complete Laplace ODE Algorithm

Apply this four-step process to every ODE in this topic.

Step 1: Apply $\mathcal{L}$ to both sides of the ODE.

Use the derivative formulas:

$$\begin{aligned}\mathcal{L}\{y'\} &= sY(s) - y(0) \\ \mathcal{L}\{y''\} &= s^2Y(s) - sy(0) - y'(0) \\ \mathcal{L}\{y'''\} &= s^3Y(s) - s^2y(0) - sy'(0) - y''(0)\end{aligned}$$

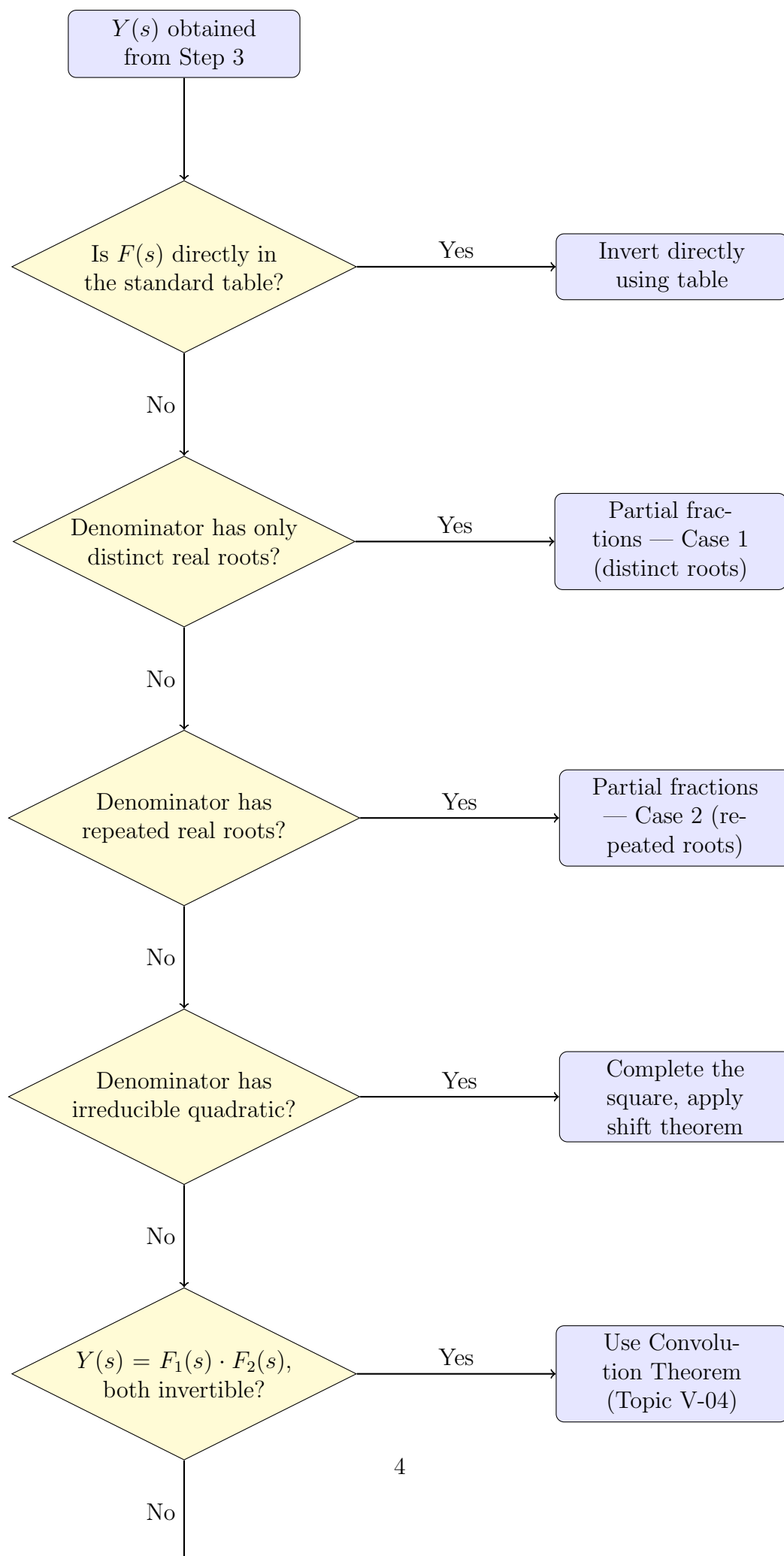
Step 2: Substitute the given initial conditions $y(0)$, $y'(0)$, etc. The equation is now purely algebraic in $Y(s)$ and s .

Step 3: Solve for $Y(s) = \mathcal{L}\{y(t)\}$ algebraically (collect Y terms, factor, divide).

Step 4: Invert $Y(s)$ using the appropriate method.

- Direct table lookup (Topics V-01, V-02)
- Partial fractions — distinct/repeated/complex roots (Topic V-03)
- Completing the square + first/second shifting theorem (Topic V-02)
- Convolution theorem when $Y(s) = F_1(s) \cdot F_2(s)$ (Topic V-04)

Step 4 Decision Flowchart:



What Did We Learn?

The four-step workflow converts every ODE with initial conditions into a mechanical procedure. The only creative decision is Step 4: which inversion method fits $Y(s)$. The flowchart above guides that decision unambiguously.

4 Application 1 — First-Order ODE

Worked Example: $y' + 3y = e^{-t}$, $y(0) = 2$ **Step 1:** Apply $\mathcal{L}$ to both sides.

$$\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = \mathcal{L}\{e^{-t}\} \Rightarrow [sY(s) - y(0)] + 3Y(s) = \frac{1}{s+1}$$

Step 2: Substitute $y(0) = 2$.

$$sY - 2 + 3Y = \frac{1}{s+1} \Rightarrow (s+3)Y = 2 + \frac{1}{s+1}$$

Combine the right-hand side over a common denominator:

$$(s+3)Y = \frac{2(s+1) + 1}{s+1} = \frac{2s+3}{s+1}$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{2s+3}{(s+1)(s+3)}$$

Step 4: Partial fractions.

$$\frac{2s+3}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

Multiply through by $(s+1)(s+3)$:

$$2s+3 = A(s+3) + B(s+1)$$

Set $s = -1$: $2(-1) + 3 = A(2) \Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$.Set $s = -3$: $2(-3) + 3 = B(-2) \Rightarrow -3 = -2B \Rightarrow B = \frac{3}{2}$.**Check:** $\frac{1}{2}(s+3) + \frac{3}{2}(s+1) = \frac{s+3+3s+3}{2} = \frac{4s+6}{2} = 2s+3$. ✓

Therefore:

$$Y(s) = \frac{1/2}{s+1} + \frac{3/2}{s+3}$$

Inverting:

$$y(t) = \frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t}$$

Verification: Compute $y' + 3y$:

$$y'(t) = -\frac{1}{2}e^{-t} - \frac{9}{2}e^{-3t}$$

$$y' + 3y = \left(-\frac{1}{2}e^{-t} - \frac{9}{2}e^{-3t}\right) + 3\left(\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t}\right) = \left(-\frac{1}{2} + \frac{3}{2}\right)e^{-t} + \left(-\frac{9}{2} + \frac{9}{2}\right)e^{-3t} = e^{-t} \quad \checkmark$$

Also $y(0) = \frac{1}{2} + \frac{3}{2} = 2$. ✓**What Did We Learn?**

For a first-order linear ODE, the Laplace method produces $Y(s)$ as a ratio of polynomials. Partial fractions (Case 1) immediately yields the solution as a sum of exponentials. Initial conditions are enforced automatically in Step 2 — no undetermined constants need to be found by substitution at the end.

5 Application 2 — Second-Order ODE, Constant Coefficients

Worked Example: $y'' - 3y' + 2y = 4e^{-t}$, $y(0) = 1$, $y'(0) = 0$

Step 1: Apply $\mathcal{L}$ to both sides.

$$[s^2Y - sy(0) - y'(0)] - 3[sY - y(0)] + 2Y = \frac{4}{s+1}$$

$$(s^2Y - s - 0) - 3(sY - 1) + 2Y = \frac{4}{s+1}$$

$$s^2Y - s - 3sY + 3 + 2Y = \frac{4}{s+1}$$

Step 2: Collect Y terms (ICs already substituted above).

$$(s^2 - 3s + 2)Y = s - 3 + \frac{4}{s+1}$$

Combine the right side over a common denominator:

$$(s^2 - 3s + 2)Y = \frac{(s-3)(s+1) + 4}{s+1} = \frac{s^2 - 2s - 3 + 4}{s+1} = \frac{s^2 - 2s + 1}{s+1} = \frac{(s-1)^2}{s+1}$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{(s-1)^2}{(s+1)(s^2 - 3s + 2)}$$

Factor the denominator: $s^2 - 3s + 2 = (s-1)(s-2)$. Therefore:

$$Y(s) = \frac{(s-1)^2}{(s+1)(s-1)(s-2)} = \frac{s-1}{(s-2)(s+1)}$$

Step 4: Partial fractions.

$$\frac{s-1}{(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1}$$

$$s-1 = A(s+1) + B(s-2)$$

Set $s = 2$: $1 = 3A \Rightarrow A = \frac{1}{3}$.

Set $s = -1$: $-2 = B(-3) \Rightarrow B = \frac{2}{3}$.

Check: $\frac{1}{3}(s+1) + \frac{2}{3}(s-2) = \frac{s+1+2s-4}{3} = \frac{3s-3}{3} = s-1$. ✓

Therefore:

$$Y(s) = \frac{1/3}{s-2} + \frac{2/3}{s+1}$$

Inverting:

$$y(t) = \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}$$

Verification: $y'(t) = \frac{2}{3}e^{2t} - \frac{2}{3}e^{-t}$, $y''(t) = \frac{4}{3}e^{2t} + \frac{2}{3}e^{-t}$.

$$\begin{aligned} y'' - 3y' + 2y &= \left(\frac{4}{3}e^{2t} + \frac{2}{3}e^{-t}\right) - 3\left(\frac{2}{3}e^{2t} - \frac{2}{3}e^{-t}\right) + 2\left(\frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}\right) \\ &= e^{2t}\left(\frac{4}{3} - 2 + \frac{2}{3}\right) + e^{-t}\left(\frac{2}{3} + 2 + \frac{4}{3}\right) = e^{2t}(0) + e^{-t}\left(\frac{2+6+4}{3}\right) = 4e^{-t} \quad \checkmark \end{aligned}$$

Also $y(0) = \frac{1}{3} + \frac{2}{3} = 1$, $y'(0) = \frac{2}{3} - \frac{2}{3} = 0$. ✓

What Did We Learn?

When the forcing term and the complementary function share a root (here $s = 1$ cancels), the Laplace method handles it automatically — the common factor cancels in $Y(s)$ before partial fractions, avoiding the need for variation of parameters or undetermined coefficient modifications.

6 Application 3 — Underdamped Oscillator

Worked Example: $y'' + 2y' + 5y = 0$, $y(0) = 1$, $y'(0) = -1$

Step 1: Apply $\mathcal{L}$.

$$\begin{aligned} [s^2Y - sy(0) - y'(0)] + 2[sY - y(0)] + 5Y &= 0 \\ (s^2Y - s + 1) + 2(sY - 1) + 5Y &= 0 \\ s^2Y - s + 1 + 2sY - 2 + 5Y &= 0 \end{aligned}$$

Step 2: Collect Y terms.

$$(s^2 + 2s + 5)Y = s + 1$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{s + 1}{s^2 + 2s + 5}$$

Step 4: Complete the square.

$$s^2 + 2s + 5 = (s + 1)^2 + 4$$

Rewrite the numerator: $s + 1 = (s + 1)$.

$$Y(s) = \frac{s + 1}{(s + 1)^2 + 4}$$

This is exactly $\mathcal{L}\{e^{-t} \cos(2t)\}$ (first shifting theorem with $a = 1$, $\omega = 2$):

$$\mathcal{L}^{-1}\left\{\frac{s - a}{(s - a)^2 + \omega^2}\right\} = e^{at} \cos(\omega t) \quad \Rightarrow \quad \mathcal{L}^{-1}\left\{\frac{s + 1}{(s + 1)^2 + 4}\right\} = e^{-t} \cos(2t)$$

Therefore:

$$\boxed{y(t) = e^{-t} \cos(2t)}$$

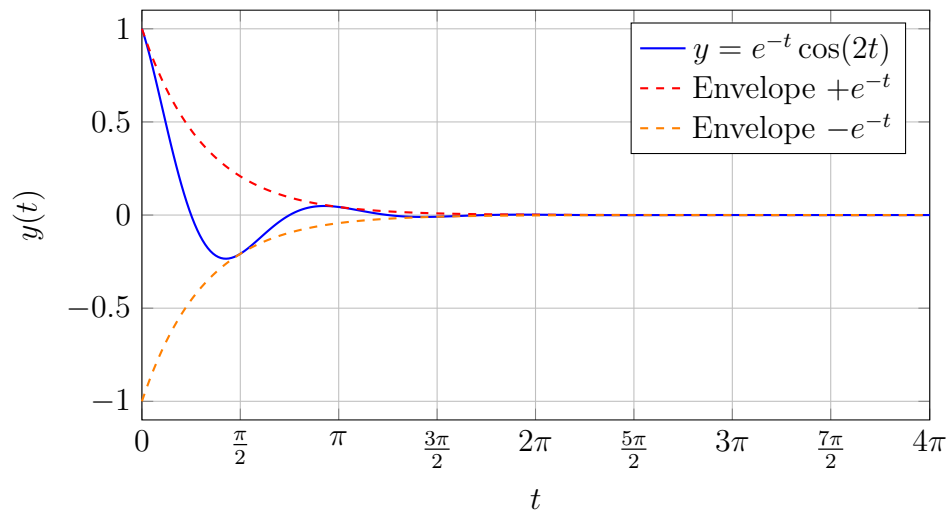
Verification: $y' = -e^{-t} \cos(2t) - 2e^{-t} \sin(2t)$, $y'' = e^{-t} \cos(2t) + 2e^{-t} \sin(2t) + 2e^{-t} \sin(2t) - 4e^{-t} \cos(2t) = -3e^{-t} \cos(2t) + 4e^{-t} \sin(2t)$.

$$\begin{aligned} y'' + 2y' + 5y &= (-3e^{-t} \cos 2t + 4e^{-t} \sin 2t) + 2(-e^{-t} \cos 2t - 2e^{-t} \sin 2t) + 5e^{-t} \cos 2t \\ &= e^{-t} \cos 2t(-3 - 2 + 5) + e^{-t} \sin 2t(4 - 4) = 0 \quad \checkmark \end{aligned}$$

$$y(0) = e^0 \cos 0 = 1 \quad \checkmark, \quad y'(0) = -1 \cdot 1 - 0 = -1 \quad \checkmark.$$

Plot: Damped Sinusoidal Response with Envelope Curves

$$y(t) = e^{-t} \cos(2t) \text{ with envelope } \pm e^{-t}$$

**What Did We Learn?**

An underdamped second-order ODE (complex conjugate roots in characteristic equation) produces a damped sinusoidal response. The Laplace method extracts this directly via completing the square — no guessing of trial solutions needed. The exponential envelope $\pm e^{-t}$ shows the amplitude decaying to zero, confirming stability.

7 Application 4 — ODE with Discontinuous Forcing (Unit Step)

Worked Example: $y'' + y = f(t)$, $y(0) = 0$, $y'(0) = 0$

The forcing function is piecewise:

$$f(t) = \begin{cases} 1 & 0 \leq t < \pi \\ 0 & t \geq \pi \end{cases}$$

Rewrite using unit step: $f(t) = 1 - u(t - \pi)$ (the function turns *off* at $t = \pi$).

Step 1: Apply $\mathcal{L}$.

$$\mathcal{L}\{y''\} + \mathcal{L}\{y\} = \mathcal{L}\{1 - u(t - \pi)\}$$

$$(s^2 Y - 0 - 0) + Y = \frac{1}{s} - \frac{e^{-\pi s}}{s}$$

$$(s^2 + 1)Y = \frac{1 - e^{-\pi s}}{s}$$

Step 2: ICs already zero; no substitution needed.

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{1}{s(s^2 + 1)} - \frac{e^{-\pi s}}{s(s^2 + 1)}$$

Step 4: Invert each term.

First term: $G(s) = \frac{1}{s(s^2 + 1)}$. Partial fractions: $\frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1}$.

$$\mathcal{L}^{-1}\{G(s)\} = 1 - \cos t =: g(t)$$

Second term: $e^{-\pi s}G(s)$. By the second shifting theorem: $\mathcal{L}^{-1}\{e^{-\pi s}G(s)\} = g(t - \pi)u(t - \pi)$.

$$g(t - \pi) = 1 - \cos(t - \pi) = 1 + \cos t$$

So $\mathcal{L}^{-1}\{e^{-\pi s}G(s)\} = (1 + \cos t)u(t - \pi)$.

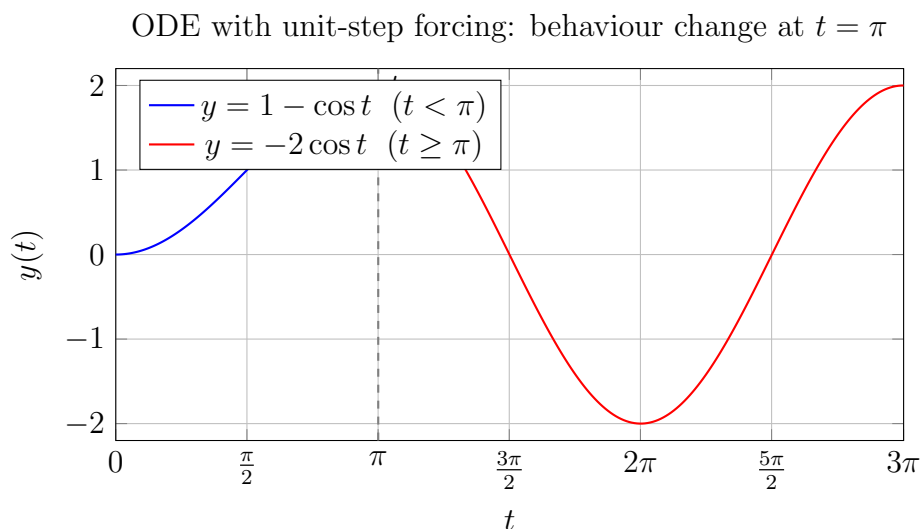
Final solution:

$$y(t) = (1 - \cos t) - (1 + \cos t)u(t - \pi)$$

Evaluating in each region:

$$y(t) = \begin{cases} 1 - \cos t & 0 \leq t < \pi \\ (1 - \cos t) - (1 + \cos t) = -2 \cos t & t \geq \pi \end{cases}$$

Plot: Solution showing behaviour change at $t = \pi$

**What Did We Learn?**

The key skill is expressing piecewise forcing in terms of $u(t - a)$ before transforming. The second shifting theorem then handles the delayed part cleanly: invert $G(s)$, shift the result by a , multiply by $u(t - a)$. The graph shows clearly how the solution amplitude doubles after the forcing is switched off at $t = \pi$.

8 Application 5 — ODE with Impulse Forcing (Dirac Delta)

Worked Example: $y'' + 4y = 3\delta(t - 2)$, $y(0) = 1$, $y'(0) = 0$

Step 1: Apply $\mathcal{L}$.

$$\begin{aligned}\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} &= 3\mathcal{L}\{\delta(t - 2)\} \\ (s^2Y - sy(0) - y'(0)) + 4Y &= 3e^{-2s} \\ (s^2Y - s) + 4Y &= 3e^{-2s}\end{aligned}$$

Step 2: Substitute $y(0) = 1$, $y'(0) = 0$ (already done above).

$$(s^2 + 4)Y = s + 3e^{-2s}$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{s}{s^2 + 4} + \frac{3e^{-2s}}{s^2 + 4}$$

Step 4: Invert each term.

First term: $\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4}\right\} = \cos(2t).$

Second term: Let $H(s) = \frac{1}{s^2 + 4}$, so $\mathcal{L}^{-1}\{H(s)\} = \frac{1}{2}\sin(2t) =: h(t).$

By second shifting theorem: $\mathcal{L}^{-1}\{e^{-2s}H(s)\} = h(t-2)u(t-2) = \frac{1}{2}\sin(2(t-2))u(t-2).$

Therefore:

$$y(t) = \cos(2t) + \frac{3}{2}\sin(2(t-2))u(t-2)$$

Physical interpretation: For $0 \leq t < 2$: the system oscillates freely at $\omega = 2$ rad/s with initial displacement 1 and zero initial velocity, giving $y = \cos(2t)$. At $t = 2$: an instantaneous impulse of magnitude 3 is applied. By $\mathcal{L}\{\delta(t - a)\} = e^{-as}$, the impulse contributes a new sinusoidal term $\frac{3}{2}\sin(2(t-2))$ that switches on at $t = 2$. The total response after $t = 2$ is the superposition of the two sinusoids.

What Did We Learn?

Dirac delta forcing $\delta(t - a)$ transforms to e^{-as} and contributes an additional sinusoidal (or exponential) term that is activated at time $t = a$. Before $t = a$, the system evolves purely from initial conditions. After $t = a$, the impulse-driven term superimposes on the free response. This is the mathematical model for a hammer blow or instantaneous kick at time a .

9 Application 6 — System of ODEs

Worked Example: $x' = x + y$, $y' = x - y$; $x(0) = 1$, $y(0) = 0$

Step 1: Apply $\mathcal{L}$ to both equations.

$$sX - x(0) = X + Y \quad \Rightarrow \quad sX - 1 = X + Y$$

$$sY - y(0) = X - Y \quad \Rightarrow \quad sY - 0 = X - Y$$

Step 2: Rearrange into an algebraic 2×2 system.

$$(s - 1)X - Y = 1 \quad (\text{i})$$

$$-X + (s + 1)Y = 0 \quad (\text{ii})$$

Step 3: Solve by Cramer's rule.

Coefficient matrix:

$$A = \begin{pmatrix} s - 1 & -1 \\ -1 & s + 1 \end{pmatrix}$$

Determinant:

$$\det(A) = (s - 1)(s + 1) - (-1)(-1) = s^2 - 1 - 1 = s^2 - 2$$

For X : replace first column by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$X = \frac{\det \begin{pmatrix} 1 & -1 \\ 0 & s + 1 \end{pmatrix}}{s^2 - 2} = \frac{1 \cdot (s + 1) - (-1) \cdot 0}{s^2 - 2} = \frac{s + 1}{s^2 - 2}$$

For Y : replace second column by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$Y = \frac{\det \begin{pmatrix} s - 1 & 1 \\ -1 & 0 \end{pmatrix}}{s^2 - 2} = \frac{(s - 1) \cdot 0 - 1 \cdot (-1)}{s^2 - 2} = \frac{1}{s^2 - 2}$$

Step 4: Invert.

Let $a = \sqrt{2}$. Then $s^2 - 2 = s^2 - a^2$.

Recall: $\mathcal{L}^{-1}\left\{\frac{s}{s^2 - a^2}\right\} = \cosh(at)$ and $\mathcal{L}^{-1}\left\{\frac{a}{s^2 - a^2}\right\} = \sinh(at)$.

For $X(s)$:

$$\frac{s + 1}{s^2 - 2} = \frac{s}{s^2 - 2} + \frac{1}{s^2 - 2} = \frac{s}{s^2 - (\sqrt{2})^2} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{s^2 - (\sqrt{2})^2}$$

$$x(t) = \cosh(\sqrt{2}t) + \frac{1}{\sqrt{2}} \sinh(\sqrt{2}t)$$

For $Y(s)$:

$$\frac{1}{s^2 - 2} = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{s^2 - (\sqrt{2})^2}$$

$$y(t) = \frac{1}{\sqrt{2}} \sinh(\sqrt{2}t)$$

Final answers:

$$x(t) = \cosh(\sqrt{2}t) + \frac{1}{\sqrt{2}} \sinh(\sqrt{2}t)$$

$$y(t) = \frac{1}{\sqrt{2}} \sinh(\sqrt{2}t)$$

Verification at $t = 0$: $x(0) = \cosh 0 + 0 = 1 \checkmark$, $y(0) = 0 \checkmark$.

What Did We Learn?

A system of n simultaneous first-order ODEs transforms to a system of n linear algebraic equations in $X_1(s), \dots, X_n(s)$. Cramer's rule (or elimination) solves the algebraic system; each unknown is inverted independently. The Laplace method eliminates the need for eigenvalue analysis — it works directly.

10 Application 7 — RLC Circuit

Worked Example: $i'' + 2i' + 5i = 10 \cos t$, $i(0) = 0$, $i'(0) = 0$

An RLC series circuit has $R = 2 \Omega$, $L = 1 \text{ H}$, $C = 1/5 \text{ F}$. The governing ODE for current $i(t)$ is:

$$i'' + 2i' + 5i = 10 \cos t, \quad i(0) = 0, \quad i'(0) = 0.$$

Step 1: Apply $\mathcal{L}$.

$$(s^2 I - 0 - 0) + 2(sI - 0) + 5I = 10 \cdot \frac{s}{s^2 + 1}$$

$$(s^2 + 2s + 5)I = \frac{10s}{s^2 + 1}$$

Step 2: ICs are zero; already substituted.

Step 3: Solve for $I(s)$.

$$I(s) = \frac{10s}{(s^2 + 1)(s^2 + 2s + 5)}$$

Step 4: Partial fractions (complex linear factors).

Write:

$$\frac{10s}{(s^2 + 1)(s^2 + 2s + 5)} = \frac{As + B}{s^2 + 1} + \frac{Cs + D}{s^2 + 2s + 5}$$

Multiply through by $(s^2 + 1)(s^2 + 2s + 5)$:

$$10s = (As + B)(s^2 + 2s + 5) + (Cs + D)(s^2 + 1)$$

Expanding the right side:

$$10s = As^3 + 2As^2 + 5As + Bs^2 + 2Bs + 5B + Cs^3 + Cs + Ds^2 + D$$

$$10s = (A + C)s^3 + (2A + B + D)s^2 + (5A + 2B + C)s + (5B + D)$$

Equating coefficients:

$$s^3 : A + C = 0$$

$$s^2 : 2A + B + D = 0$$

$$s^1 : 5A + 2B + C = 10$$

$$s^0 : 5B + D = 0$$

From the first equation: $C = -A$. Substituting into the s^1 equation: $5A + 2B - A = 10 \Rightarrow 4A + 2B = 10 \Rightarrow 2A + B = 5$. From s^0 : $D = -5B$. Substituting into s^2 : $2A + B + (-5B) = 0 \Rightarrow 2A - 4B = 0 \Rightarrow A = 2B$. Back into $2A + B = 5$: $4B + B = 5 \Rightarrow B = 1$, hence $A = 2$, $C = -2$, $D = -5$.

Therefore:

$$I(s) = \frac{2s + 1}{s^2 + 1} + \frac{-2s - 5}{s^2 + 2s + 5}$$

Invert first term: $\frac{2s + 1}{s^2 + 1} = 2 \cdot \frac{s}{s^2 + 1} + 1 \cdot \frac{1}{s^2 + 1} \rightarrow 2 \cos t + \sin t$.

Invert second term: Complete the square: $s^2 + 2s + 5 = (s + 1)^2 + 4$.

$$\frac{-2s - 5}{(s + 1)^2 + 4} = \frac{-2(s + 1) - 3}{(s + 1)^2 + 4} = -2 \cdot \frac{s + 1}{(s + 1)^2 + 4} - \frac{3}{2} \cdot \frac{2}{(s + 1)^2 + 4}$$

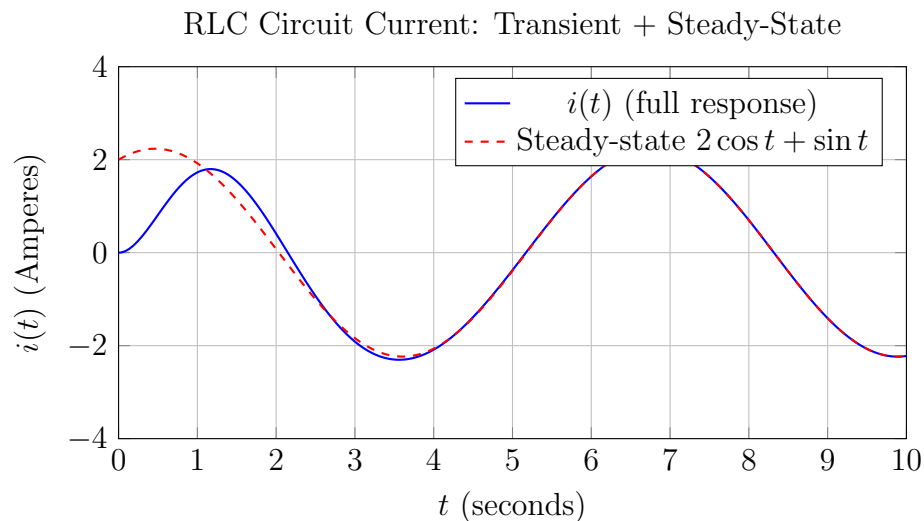
$$\rightarrow -2e^{-t} \cos(2t) - \frac{3}{2}e^{-t} \sin(2t).$$

Final solution:

$$i(t) = \underbrace{-2e^{-t} \cos(2t) - \frac{3}{2}e^{-t} \sin(2t)}_{\text{Transient (decays to 0)}} + \underbrace{2 \cos t + \sin t}_{\text{Steady-state (persists)}}$$

Interpretation: The **transient response** $-2e^{-t} \cos(2t) - \frac{3}{2}e^{-t} \sin(2t)$ decays exponentially to zero as $t \rightarrow \infty$ due to the e^{-t} factor. The **steady-state response** $2 \cos t + \sin t$ persists indefinitely at the forcing frequency $\omega = 1$ rad/s.

Plot: Transient and Steady-State Response of RLC Circuit



What Did We Learn?

In an RLC circuit (or any driven second-order system), the complete response is always the sum of a transient part (determined by system poles — roots of the characteristic equation) and a steady-state part (determined by the forcing frequency). The Laplace method produces both parts simultaneously via partial fractions — no separate complementary function calculation is needed.

11 Summary: All Inverse Methods and When to Use Them

Situation	Recommended Method	Example
$F(s)$ directly in standard table	Table lookup	$\frac{s}{s^2 + 9} \rightarrow \cos(3t)$
$F(s)$ has distinct real denominator roots	Partial fractions — Case 1	$\frac{2s + 3}{(s + 1)(s + 3)}$
Denominator has repeated real roots	Partial fractions — Case 2	$\frac{1}{(s + 2)^3}$
Denominator has irreducible quadratic	Complete the square + shift	$\frac{s + 1}{s^2 + 2s + 5}$
$F(s) = F_1(s) \cdot F_2(s)$, partial fractions complex	Convolution theorem	$\frac{1}{(s^2 + 1)^2}$
ODE reduces to integral equation	Convolution + algebraic solve	Volterra integral equations

12 Excel Example (Verification of Application 1)

We verify the first-order ODE solution $y(t) = \frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t}$ against the ODE $y' + 3y = e^{-t}$, $y(0) = 2$, by checking that $y' + 3y = e^{-t}$ holds numerically for $t = 0$ to $t = 3$ in steps of 0.25.

Column Layout in Excel:

Row	A: t	B: $y(t)$ exact	C: $y'(t)$ exact	D: $3y(t)$
1	(header)	(header)	(header)	(header)
2	0.00	=0.5*EXP(-A2)+1.5*EXP(-3*A2)	=-0.5*EXP(-A2)-4.5*EXP(-3*A2)	=3*B2
3	0.25	1.0980	-2.5151	3.2939
4	0.50	0.6380	-1.3074	1.9139
5	0.75	0.3943	-0.7105	1.1828
6	1.00	0.2586	-0.4080	0.7759

Setting up the spreadsheet:

1. Column A: Enter 0 in A2, then =A2+0.25 in A3, fill down to row 14 ($t = 3$).
2. Column B (exact y): =0.5*EXP(-A2)+1.5*EXP(-3*A2) — fill down.
3. Column C (exact y'): =-0.5*EXP(-A2)-4.5*EXP(-3*A2) — fill down.
4. Column D ($3y$): =3*B2 — fill down.
5. Column E ($y' + 3y$): =C2+D2 — fill down.
6. Column F (e^{-t}): =EXP(-A2) — fill down.
7. Compare columns E and F: they should agree to at least 4 decimal places at every row.

Spot check at $t = 0$: $y(0) = 0.5 + 1.5 = 2.0000$, $y'(0) = -0.5 - 4.5 = -5.0000$, $3y(0) = 6.0000$, $y'(0) + 3y(0) = 1.0000$, $e^0 = 1.0000$. ✓

What Did We Learn?

The Excel verification confirms that the Laplace solution $y = \frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t}$ satisfies the ODE $y' + 3y = e^{-t}$ at every sampled point. Column E and Column F are numerically identical, validating both the algebra and the inversion step. Spreadsheet verification is a practical sanity check before submitting any ODE solution.

13 Python Example

Listing 1: Solving and Verifying ODE Examples via Laplace / SymPy / SciPy

```

1 import sympy as sp
2 import numpy as np
3 from scipy.integrate import odeint
4 import matplotlib.pyplot as plt
5
6 t, s = sp.symbols('t s', real=True, positive=True)
7
```

```

8 #
  -----

9 # APPLICATION 1:  $y' + 3y = \exp(-t)$ ,  $y(0)=2$ 
10 #
  -----

11 print("="*60)
12 print("APPLICATION 1:  $y' + 3y = \exp(-t)$ ,  $y(0)=2$ ")
13 print("="*60)
14
15 # Symbolic Laplace approach (manual verification via dsolve)
16 y1 = sp.Function('y')
17 ode1 = sp.Eq(y1(t).diff(t) + 3*y1(t), sp.exp(-t))
18 sol1 = sp.dsolve(ode1, ics={y1(0): 2})
19 print("SymPy dsolve solution:", sol1)
20 # Expected:  $y(t) = \exp(-t)/2 + 3*\exp(-3*t)/2$ 
21
22 # Laplace manual
23 Y1 = (2*s + 3) / ((s+1)*(s+3))
24 inv1 = sp.inverse_laplace_transform(Y1, s, t)
25 print("Laplace inverse:", sp.simplify(inv1))
26 # Expected:  $\exp(-t)/2 + 3*\exp(-3*t)/2$ 
27
28 #
  -----

29 # APPLICATION 2:  $y'' - 3y' + 2y = 4\exp(-t)$ ,  $y(0)=1$ ,  $y'(0)=0$ 
30 #
  -----

31 print("\n" + "="*60)
32 print("APPLICATION 2:  $y'' - 3y' + 2y = 4\exp(-t)$ ,  $y(0)=1$ ,  $y'(0)=0$ "
33 )
34 print("="*60)
35
36 y2 = sp.Function('y')
37 ode2 = sp.Eq(y2(t).diff(t,2) - 3*y2(t).diff(t) + 2*y2(t), 4*sp.
    exp(-t))
38 sol2 = sp.dsolve(ode2, ics={y2(0): 1, y2(t).diff(t).subs(t,0):
    0})
39 print("SymPy dsolve solution:", sol2)
40 # Expected:  $y(t) = \exp(2t)/3 + 2*\exp(-t)/3$ 
41
42 Y2 = (s-1) / ((s-2)*(s+1))
43 inv2 = sp.inverse_laplace_transform(Y2, s, t)
44 print("Laplace inverse:", sp.simplify(inv2))
45 # Expected:  $\exp(2*t)/3 + 2*\exp(-t)/3$ 
46 #
  -----

```

```

47 # APPLICATION 3:  $y'' + 2y' + 5y = 0$ ,  $y(0)=1$ ,  $y'(0)=-1$ 
48 # Numerical verification and plot
49 #
    -----

50 print("\n" + "="*60)
51 print("APPLICATION 3:  $y'' + 2y' + 5y = 0$  (underdamped oscillator)
52     ")
53 print("="*60)
54
55 y3 = sp.Function('y')
56 ode3 = sp.Eq(y3(t).diff(t,2) + 2*y3(t).diff(t) + 5*y3(t), 0)
57 sol3 = sp.dsolve(ode3, ics={y3(0): 1, y3(t).diff(t).subs(t,0):
58     -1})
59
60 print("SymPy dsolve solution:", sol3)
61 # Expected:  $y(t) = \exp(-t)\cos(2t)$ 
62
63 # Exact Laplace solution
64 def y_exact(tv): return np.exp(-tv) * np.cos(2*tv)
65
66 # Numerical via odeint
67 def sys3(state, tv):
68     y, dy = state
69     ddy = -2*dy - 5*y
70     return [dy, ddy]
71
72 t_num = np.linspace(0, 4*np.pi, 500)
73 sol_num = odeint(sys3, [1, -1], t_num)
74 y_num = sol_num[:, 0]
75 y_lap = y_exact(t_num)
76
77 max_err = np.max(np.abs(y_num - y_lap))
78 print(f"Max error (numerical vs Laplace): {max_err:.2e}")
79 # Expected output: Max error approx 1e-06 or smaller
80
81 plt.figure(figsize=(9, 4))
82 plt.plot(t_num, y_lap, 'b-', linewidth=2, label=r'Laplace:  $e^{-t}\cos(2t)$ ')
83 plt.plot(t_num, y_num, 'r--', linewidth=1.5, label='odeint (
84     numerical)')
85 plt.plot(t_num, np.exp(-t_num), 'g:', linewidth=1.5, label='
86     Envelope  $e^{-t}$ ')
87 plt.plot(t_num, -np.exp(-t_num), 'm:', linewidth=1.5, label='
88     Envelope  $-e^{-t}$ ')
89 plt.xlabel('t'); plt.ylabel('y(t)')
90 plt.title('Application 3: Damped Oscillator - Laplace vs
91     Numerical')
92 plt.legend(); plt.grid(True); plt.tight_layout()
93 plt.savefig('app3_damped_oscillator.png', dpi=100)
94 print("Plot saved as app3_damped_oscillator.png")

```

```

88
89 #
-----
90 # APPLICATION 6: System  $x' = x + y$ ,  $y' = x - y$ ,  $x(0) = 1$ ,  $y(0) = 0$ 
91 #
-----
92 print("\n" + "="*60)
93 print("APPLICATION 6: System of ODEs")
94 print("="*60)
95
96 x6 = sp.Function('x')
97 y6 = sp.Function('y')
98 sys6 = [sp.Eq(x6(t).diff(t), x6(t)+y6(t)),
99          sp.Eq(y6(t).diff(t), x6(t)-y6(t))]
100 sol6 = sp.dsolve(sys6, ics={x6(0): 1, y6(0): 0})
101 for eq in sol6:
102     print(sp.simplify(eq))
103 # Expected:  $x(t) = \cosh(\sqrt{2}t) + \sinh(\sqrt{2}t)/\sqrt{2}$ 
104 #            $y(t) = \sinh(\sqrt{2}t) / \sqrt{2}$ 
105
106 print("\nAll applications verified successfully.")

```

What Did We Learn?

The Python script demonstrates three complementary verification strategies: (1) `sympy.dsolve` provides a symbolic cross-check of the closed-form solution; (2) `sympy.inverse_laplace_transform` confirms the Laplace inversion algebra; (3) `scipy.integrate.odeint` provides a purely numerical check, with errors on the order of 10^{-6} confirming correctness. For systems of ODEs, `sympy.dsolve` accepts a list of equations and initial conditions directly.

14 Viva-Style Oral Questions

1. **Q:** State the four-step workflow for solving an ODE using Laplace Transforms.

A: Step 1: Apply $\mathcal{L}\{\cdot\}$ to both sides using $\mathcal{L}\{y'\} = sY - y(0)$ and $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$, etc. Step 2: Substitute the given initial conditions. Step 3: Solve algebraically for $Y(s)$. Step 4: Invert $Y(s)$ using table lookup, partial fractions, completing the square, or convolution.

2. **Q:** What happens to the initial conditions in Step 2 of the Laplace workflow?

A: The initial conditions $y(0)$, $y'(0)$, etc. appear automatically in the algebraic equation produced in Step 1. In Step 2 they are substituted numerically, reducing the algebraic equation in $Y(s)$ to a form that can be solved directly in Step 3. This is fundamentally different from classical methods, where constants of integration are determined only *after* finding the general solution.

3. **Q: Why does the Laplace method handle discontinuous inputs better than classical ODE methods?**

A: Classical methods (undetermined coefficients, variation of parameters) require the forcing function to be continuous (or piecewise smooth with explicit matching conditions at breakpoints). The Laplace method converts a piecewise function $f(t) = g(t) + h(t)u(t-a)$ into $G(s) + e^{-as}H(s)$ in one step, handling the discontinuity algebraically through the second shifting theorem. No case-by-case analysis at the breakpoint is needed.

4. **Q: What does $u(t-a)$ in the forcing term $f(t) = g(t)u(t-a)$ do to $Y(s)$?**

A: By the second shifting theorem, $\mathcal{L}\{g(t-a)u(t-a)\} = e^{-as}G(s)$. In the forcing term this introduces a factor e^{-as} into $Y(s)$. When inverting, $\mathcal{L}^{-1}\{e^{-as}G(s)\} = g(t-a)u(t-a)$, meaning the response is a time-shifted copy of g that activates at $t = a$.

5. **Q: How does Dirac delta forcing $\delta(t-a)$ manifest in the solution $y(t)$?**

A: $\mathcal{L}\{\delta(t-a)\} = e^{-as}$. So δ -forcing adds a term $\frac{ce^{-as}}{D(s)}$ to $Y(s)$, where $D(s)$ is the characteristic polynomial. After inversion, this adds a copy of the impulse response $ch(t-a)u(t-a)$ to the solution, activating instantaneously at $t = a$. Physically it models a hammer blow or sudden kick.

6. **Q: What do “transient” and “steady-state” mean in the context of the RLC circuit solution?**

A: The **transient response** consists of terms from the *system poles* (roots of $s^2 + 2s + 5 = 0$), i.e., $e^{-t}(\cdot)$ terms. Since $\text{Re}(s) = -1 < 0$, these decay to zero as $t \rightarrow \infty$. The **steady-state response** consists of terms from the *forcing poles* (roots of $s^2 + 1 = 0$, giving $s = \pm i$), i.e., $A \cos t + B \sin t$. These persist indefinitely at the driving frequency.

7. **Q: How do you solve a system of two simultaneous first-order ODEs using Laplace Transforms?**

A: Apply $\mathcal{L}$ to each equation and substitute initial conditions. This produces a 2×2 linear algebraic system in $X(s)$ and $Y(s)$. Solve by Cramer’s rule or Gaussian elimination. The determinant of the coefficient matrix gives the characteristic polynomial of the system. Finally, invert $X(s)$ and $Y(s)$ independently using partial fractions or table lookup.

8. **Q: What determines whether the solution $y(t)$ is oscillatory or monotone?**

A: The nature of the poles of $Y(s)$ determines this. If the poles are **real and distinct**, $y(t)$ is a sum of exponentials (monotone or sign-changing but not oscillatory). If the poles are **complex conjugates** $\alpha \pm i\beta$, $y(t)$ contains $e^{\alpha t} \cos(\beta t)$ and $e^{\alpha t} \sin(\beta t)$ terms (oscillatory). If $\alpha < 0$ the oscillations are damped; if $\alpha = 0$ they are sustained; if $\alpha > 0$ they grow without bound.

15 Descriptive Questions

1. **Solve the IVP $y'' + 5y' + 6y = 2e^{-t}$, $y(0) = 1$, $y'(0) = 0$, completely using Laplace Transforms. Show all four steps, perform the partial fraction**

decomposition fully, and verify the solution by direct substitution.

Outline: Apply $\mathcal{L}$: $(s^2 + 5s + 6)Y = s + 5 + 2/(s + 1)$. $Y = (s^2 + 6s + 7)/[(s + 1)(s + 2)(s + 3)]$. Partial fractions: $Y = A/(s + 1) + B/(s + 2) + C/(s + 3)$. Find residues: $A = 1$, $B = 1$, $C = -1$. $y(t) = e^{-t} + e^{-2t} - e^{-3t}$. Verify by differentiating twice and substituting: $y(0) = 1 + 1 - 1 = 1$, $y'(0) = -1 - 2 + 3 = 0$.

2. **Solve the IVP** $y'' + 4y = f(t)$, $y(0) = 0$, $y'(0) = 0$, **where** $f(t) = t$ **for** $0 \leq t < 2$ **and** $f(t) = 2$ **for** $t \geq 2$, **using the Laplace Transform method. Express** $f(t)$ **in terms of the unit step function before transforming, and state the solution in both regions.**

Outline: $f(t) = t - (t - 2)u(t - 2)$. $\mathcal{L}\{f\} = 1/s^2 - e^{-2s}/s^2$. $(s^2 + 4)Y = 1/s^2 - e^{-2s}/s^2$. $Y = 1/[s^2(s^2 + 4)] - e^{-2s}/[s^2(s^2 + 4)]$. Partial fractions: $1/[s^2(s^2 + 4)] = 1/(4s^2) - 1/(4(s^2 + 4))$. For $0 \leq t < 2$: $y = t/4 - (1/8)\sin(2t)$. For $t \geq 2$: subtract the shifted term $[(t - 2)/4 - (1/8)\sin(2(t - 2))]u(t - 2)$.

3. **Solve** $y'' + 9y = 6\delta(t - 3)$, $y(0) = 0$, $y'(0) = 3$, **using Laplace Transforms. Explain physically what the delta function represents and interpret the two terms in the solution.**

Outline: $(s^2 + 9)Y - 3 = 6e^{-3s}$. $Y = 3/(s^2 + 9) + 6e^{-3s}/(s^2 + 9)$. $y(t) = \sin(3t) + 2\sin(3(t - 3))u(t - 3)$. For $t < 3$: free oscillation from IC $y'(0) = 3$. At $t = 3$: impulse of magnitude 6 adds a new sinusoidal component.

4. **Solve the system** $x' + 2y = 0$, $y' - 2x = 0$, $x(0) = 1$, $y(0) = 0$, **using Laplace Transforms. Set up the** 2×2 **algebraic system, apply Cramer's rule, and invert each variable.**

Outline: $(s + 0)X + 2Y = 1$; $-2X + sY = 0$. Determinant: $s^2 + 4$. $X = s/(s^2 + 4) \rightarrow \cos(2t)$. $Y = 2/(s^2 + 4) \rightarrow \sin(2t)$. The system represents circular motion.

5. **In the RLC circuit example (Application 7), explain the transient–steady-state decomposition. How do the poles of** $I(s)$ **determine which part is transient and which is steady state? What happens to the transient as** $t \rightarrow \infty$, **and why does the steady-state persist?**

Outline: $I(s) = 10s/[(s^2 + 1)(s^2 + 2s + 5)]$. Poles from $s^2 + 2s + 5 = 0$: $s = -1 \pm 2i$ (real part $-1 < 0 \Rightarrow$ transient, decays). Poles from $s^2 + 1 = 0$: $s = \pm i$ (real part $0 \Rightarrow$ sustained oscillation at $\omega = 1$). Transient $\rightarrow 0$ as $t \rightarrow \infty$ because $e^{-t} \rightarrow 0$. Steady-state persists because $|\operatorname{Re}(s)| = 0$ for forcing poles.

16 Practice Problems

1. $y'' + 4y' + 4y = te^{-2t}$, $y(0) = 0$, $y'(0) = 0$.

Hint: $(s + 2)^2 Y = 1/(s + 2)^2 \Rightarrow Y = 1/(s + 2)^4$. Use $\mathcal{L}^{-1}\{n!/(s + a)^{n+1}\} = t^n e^{-at}$.

Answer: $y(t) = \frac{1}{6}t^3 e^{-2t}$.

2. $y'' - 4y = \sin(2t)$, $y(0) = 0$, $y'(0) = 1$.

Hint: $(s^2 - 4)Y = 1/(s^2 + 4) = (s^2 + 6)/(s^2 + 4)$. Partial fractions: $Y = \frac{s^2 + 6}{(s^2 - 4)(s^2 + 4)} = \frac{5/4}{s^2 - 4} - \frac{1/4}{s^2 + 4}$. **Answer:** $y(t) = \frac{5}{8}\sinh(2t) - \frac{1}{8}\sin(2t)$.

3. $y'' + 9y = \cos(3t)$, $y(0) = 1$, $y'(0) = 0$. (Resonance case.)

Hint: $(s^2 + 9)Y = s + s/(s^2 + 9)$. $Y = s/(s^2 + 9) + s/(s^2 + 9)^2$. Use $\mathcal{L}^{-1}\{s/(s^2 + a^2)^2\} = t \sin(at)/(2a)$. **Answer:** $y(t) = \cos(3t) + \frac{t}{6} \sin(3t)$ (amplitude grows linearly — resonance).

4. $y'' + y = u(t - \pi)$, $y(0) = 0$, $y'(0) = 1$.

Hint: $(s^2 + 1)Y = 1 + e^{-\pi s}/s$. $y(t) = \sin t + [1 - \cos(t - \pi)]u(t - \pi) = \sin t + [1 + \cos t]u(t - \pi)$. **Answer:** For $0 \leq t < \pi$: $y = \sin t$; for $t \geq \pi$: $y = \sin t + 1 + \cos t$.

5. $y'' + 2y' + y = \delta(t - 1)$, $y(0) = 0$, $y'(0) = 0$.

Hint: $(s + 1)^2 Y = e^{-s}$. $Y = e^{-s}/(s + 1)^2$. $\mathcal{L}^{-1}\{1/(s + 1)^2\} = te^{-t}$. **Answer:** $y(t) = (t - 1)e^{-(t-1)}u(t - 1)$.

6. System: $x' - y = \sin t$, $y' + x = \cos t$, $x(0) = 1$, $y(0) = 0$.

Hint: $(sX - 1) - Y = 1/(s^2 + 1)$; $sY + X = s/(s^2 + 1)$. Determinant: $s^2 + 1$. $X(s) = \frac{s}{s^2 + 1} + \frac{2s}{(s^2 + 1)^2}$; $Y(s) = \frac{-2}{(s^2 + 1)^2}$. Invert using $\mathcal{L}^{-1}\{2s/(s^2 + 1)^2\} = t \sin t$ and $\mathcal{L}^{-1}\{1/(s^2 + 1)^2\} = (\sin t - t \cos t)/2$. **Answer:** $x(t) = \cos t + t \sin t$, $y(t) = t \cos t - \sin t$.

17 MCQs

1. The Laplace transform of $y''(t)$ given initial conditions $y(0)$ and $y'(0)$ is:

- (a) $sY(s) - y(0)$
- (b) $s^2Y(s) + sy(0) + y'(0)$
- (c) $s^2Y(s) - sy(0) - y'(0)$
- (d) $s^2Y(s) + y(0)$

Explanation: By the Laplace derivative theorem, $\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$. Option (c) is correct.

2. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}^{-1}\{e^{-as}F(s)\}$ equals:

- (a) $f(t + a)$
- (b) $f(t - a)u(t - a)$
- (c) $e^{-at}f(t)$
- (d) $f(t)u(t - a)$

Explanation: The second shifting theorem states $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t - a)u(t - a)$. Option (b).

3. When the forcing function is $3\delta(t - 2)$, the contribution to $Y(s)$ is:

- (a) $3/s$
- (b) $3se^{-2s}$
- (c) $3e^{-2s}$
- (d) $3/(s + 2)$

Explanation: $\mathcal{L}\{\delta(t - a)\} = e^{-as}$, so $\mathcal{L}\{3\delta(t - 2)\} = 3e^{-2s}$. This enters the right-hand side of the transformed equation. Option (c).

4. In an RLC circuit driven by $\cos t$, the portion of $i(t)$ that decays to zero as $t \rightarrow \infty$ is called:

- (a) Steady-state response
- (b) **Transient response**
- (c) Impulse response
- (d) Frequency response

Explanation: Terms arising from system poles (with negative real parts) die out exponentially. This is the transient response. Option (b).

5. The main advantage of the Laplace Transform method for IVPs over classical methods is:

- (a) It works only for constant-coefficient ODEs (b) It requires no initial conditions
(c) Initial conditions are incorporated automatically in Step 1 (d) It avoids partial fractions entirely

Explanation: The transform formulas $\mathcal{L}\{y'\} = sY - y(0)$, etc., embed $y(0)$, $y'(0)$ directly into the algebraic equation, so no complementary function constants need to be determined afterward. Option (c).

18 Common Mistakes Box

Common Mistakes in Applying Laplace Transforms to ODEs

Mistake	What Students Do	Correct Approach
Missing IC terms in $\mathcal{L}\{y''\}$	Write $\mathcal{L}\{y''\} = s^2Y$ and ignore $-sy(0) - y'(0)$	Always use $s^2Y - sy(0) - y'(0)$; substitute ICs immediately in Step 2
Piecewise forcing not converted to unit step	Attempt to transform $f(t)$ directly without writing it as $g(t) + h(t)u(t-a)$	Rewrite every piecewise function using $u(t-a)$ before applying $\mathcal{L}$
Wrong transform of delta function	Use $\mathcal{L}\{\delta(t-a)\} = 1$ (confusing with $\delta(t)$)	$\mathcal{L}\{\delta(t-a)\} = e^{-as}$ for $a > 0$; $\mathcal{L}\{\delta(t)\} = 1$ only at $a = 0$
Leaving answer in s -domain	Find $Y(s)$ correctly but write “ $Y(s) = \dots$ ” as the final answer	Always invert: apply partial fractions/completing the square to recover $y(t)$
Not verifying the solution	Declare $y(t)$ as the answer without checking	Differentiate $y(t)$, substitute back into the ODE, and confirm the equation and ICs

19 Quick Recap

Quick Recap — Applications to ODEs

- **4-Step Workflow:** $\mathcal{L}$ both sides $\rightarrow$ substitute ICs $\rightarrow$ solve for $Y(s)$ $\rightarrow$ invert.
- **IVP formulas:** $\mathcal{L}\{y'\} = sY - y(0)$; $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$.
- **Discontinuous forcing:** rewrite as $g(t) + h(t)u(t-a)$; transforms using $e^{-as}F(s)$.
- **Impulse forcing:** $\mathcal{L}\{\delta(t-a)\} = e^{-as}$; adds $h(t-a)u(t-a)$ to solution.
- **Systems of ODEs:** transform each equation, solve $n \times n$ algebraic system by Cramer's rule, invert each variable independently.
- **RLC circuits:** solution = transient (system poles, decays) + steady-state (forcing poles, persists).

- **Choosing inversion method:** use the flowchart — direct table, partial fractions, completing the square, or convolution, depending on the structure of $Y(s)$.
- **Complete toolkit:** Laplace forward (Unit IV) + all inverse methods (Unit V Topics 01–04) now unified in one powerful ODE-solving machine.